PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis
AUTHORS	Amaral, Julio Mario; Kimergård, Andreas; Deluca, Paolo

VERSION 1 – REVIEW

REVIEWER	Fink, Julius Juntendo University
REVIEW RETURNED	27-Aug-2021

GENERAL COMMENTS	The authors investigated an interesting topic. This is a well-conducted meta-analysis, however, I have some concerns I'd like the authors to address: Abstract What is the main reason for AAS abusers to be reluctant to seek medical advice? Is it the legal status of AAS? Please extrapolate. Introduction The introduction is too superficial and needs more depth. Especially with regard to the effects of AAS. Many AAS have been developed to treat diseases and have several benefits. Also, the side effects should be explained in more detail and it should be emphasized that the abuse of AAS has the potential to lead to those side effects. Testosterone is very important in order to treat hypogonadism and has been shown crucial for healthy aging in men with regard to glucose/fat metabolism, cardiovascular, bone and liver health. It is important that the reader understands that most AAS are medications intended to treat diseases and if abused, like any other substance, they can lead to side effects. I would suggest to add information and refer to: Fink, Julius, Masahito Matsumoto, and Yoshifumi Tamura. "Potential application of testosterone replacement therapy as treatment for obesity and type 2 diabetes in men." Steroids 138 (2018): 161-166. Fink, Julius E., et al. "Mobility and biomechanical functions in the aging male: testosterone and the locomotive syndrome." The Aging Male (2018). Also, it is important to point out that many side effects of AAS abuse may arise from the way of procurement, with many drugs being produced under non-controlled environments. Also, the fact that AAS are, for the most part, controlled substances and that AAS abusers often buy them via illegal ways might be the main reason why AAS abusers are reluctant to seek medical advice. Please refer to: Fink, Julius, et al. "Anabolic-androgenic steroids: procurement and administration practices of doping athletes." The Physician and sportsmedicine 47.1 (2019): 10-14. Discussion and conclusion The authors do not point out why AAS abusers are reluctant to seek medical help. The main reason being the fact that the way of procurement of AAS is often illegal in many countries is probably the

	cause. If AAS could be legally purchased over the counter at every pharmacy, users would not be afraid to seek medical advice. Methods It is very important to include the legal status of AAS of each country in the analysis. Are AAS abusers more open to seeking medical advice in countries where the legislation with regard to AAS is not as strict? Please add this parameter in the meta-analysis. Also, do the authors know what type of medical advice/consultation AAS abusers were looking for? Is it for the treatment of side effects? Or is it to monitor their overall health?

REVIEWER	Boardley, Ian University of Birmingham, Statistics, rehabilitation
REVIEW RETURNED	29-Sep-2021

GENERAL COMMENTS	Many thanks for the opportunity to review this manuscript, which I believe should be of interest to readers of BMJ Open. Overall, I think it reports a nice piece of work that has the potential to make a useful and interesting contribution to the literature. In the following specific comments, I detail opportunities to further strengthen the manuscript and bring it up to a publishable standard if addressed. 1. The manuscript would benefit from a careful proof reading, as there are numerous grammatical errors present (e.g., missing the word 'the' prior to the number of studies identified). 2. P.5, l. 45-49. Please also include the secondary aims here too, as done in the abstract. The study rationale in the intro should incorporate the rationale for the secondary aims too. 3. Table 1. The exclusion criteria should include studies that intentionally recruited participants from a sub-population that had an increased or decreased likelihood of engaging with physicians. For example, did any studies recruit only participants that engaged with harm reduction services. As shown by the study findings, this group are more likely to consult with a physician than AAS users who do not use harm reduction services. This is therefore a potential source of bias that could skew the estimates produced in the meta-analysis if such studies have been included. I could not see this assessment represented within the exclusion criteria. 4. P.8, l.20-26. This is an example of an aspect of the study that is not incorporated in the aim stated at the end of the introduction. The aims need to include all aspects of the study, and these aspects need to be incorporated in the study rationale established in the introduction. 5. P.8, l.43. Study reference 53 is listed as a study from Pakistan in the reference list, but South Africa is stated here. 6. P.12, l.17-19. Was there a reason for selecting the year 2000, or was this an arbitrary selection? Given the low number of studies prior to 2000, I am not convinced this sub-analysis adds much. 7. P.13, l.40-56. I am not sure there is sufficient sample sizes to support the majority of these comparisons. 8. P.14, l.16-39. Same issue here. If these comparisons are retained, the limitation regarding sub-cell sizes should be clearly acknowledged. They may be interesting, but they cannot be relied upon. 9. P.17, l.44-59. It would be good to see some discussion of why ranges varied so widely in most countries. Presently, this paragraph mostly just expands on the relevant results rather than discussing why this may be the case. Likewise, what might explain the higher and less diverse prevalence rates in Australia compared to other countries/regions?

REVIEWER	Lewin, Adriane UNSW, Medicine/Whitlam Orthopaedic Research Centre
REVIEW RETURNED	12-Jan-2022

GENERAL COMMENTS	I enjoyed reading your paper and appreciate the amount of work that must have gone into the project so far. Please note: I checked 'major revisions' but really all the information is here already and the paper just needs a major reorganisation. It is a good paper and the main findings are likely interesting and relevant to the field of research. For these reasons it is important to ensure the results are clear, concise and easy to understand, which will improve accessibility and therefore - possibly - impact.Please see attached file for detailed comments.

VERSION 1 – AUTHOR RESPONSE

	Reviewer 1: Dr Julius Fink	
	Abstract What is the main reason for AAS abusers to be reluctant to seek medical advice? Is it the legal status of AAS? Please extrapolate.	A discussion about potential reasons influencing the engagement of AAS with physicians was added to the study, including the legal status of AAS. This discussion is now mentioned in the abstract.
	The introduction is too superficial and needs more depth. Especially with regard to the effects of AAS.	The introduction was expanded, including further description of the effects of AAS.
	It is important that the reader understands that most AAS are medications intended to treat diseases and if abused, like any other substance, they can lead to side effects. I would suggest to add information and refer to Fink et al, 2018.	This information was included in the introduction and discussion of results, including the suggested reference.
	It is important to point out that many side effects of AAS abuse may arise from the way of procurement, with many drugs being produced under non-controlled environments. Also, the fact that AAS are, for the most part, controlled substances and that AAS abusers often buy them via illegal ways might be the main reason why AAS abusers are reluctant to seek medical advice. Please refer to Fink et al, 2019.	A discussion about the impact of the legal status was included in the introduction and discussion, including the suggested reference. We also discuss how the legal status of AAS might lead some AAS users to seek substances and advice from informal sources.
	Discussion and conclusion The authors do not point out why AAS abusers are reluctant to seek medical help.	This discussion was added to the study.

	Methods It is very important to include the legal status of AAS of each country in the analysis. Please add this parameter in the meta-analysis.	We included a discussion about the legal status of AAS in each of the countries represented by studies in the review, as well as links to their legal instruments, when available. We chose not to include the legal status of AAS as a new subgroup/moderator for meta-analysis, but this comparison was included in the discussion.
	Do the authors know what type of medical advice/consultation AAS abusers were looking for?	We mention the type of support sought by AAS users as described in the studies. We considered that further explorations of the types of support sought would be beyond the goals of this study, as they were recently described by other reviews, as cited in the text.
	Reviewer 2: Dr Ian Boardley	
	The manuscript would benefit from a careful proof reading	The proof reading was performed and the grammatical errors were corrected.
	Include the secondary aims in the abstract and in the study's rationale in the introduction.	The secondary arms were included in the abstract and in study's rationale in the introduction.
	The exclusion criteria should include studies that intentionally recruited participants from a sub- population that had an increased or decreased likelihood of engaging with physicians. For example, did any studies recruit only participants that engaged with harm reduction services. As shown by the study findings, this group are more likely to consult with a physician than AAS users who do not use harm reduction services. This is therefore a potential source of bias that could skew the estimates produced in the meta-analysis if such studies have been included. I could not see this assessment represented within	As a subpopulation, AAS users recruited in the NSP have shown, as expected, a higher prevalence of seeking support from physicians. The individual prevalence rate of each NSP study, however, was not higher than many other studies with nonspecific subpopulations of AAS users, and a meta-regression did not find a statistically significant impact of the prevalence of NSP studies on the overall prevalence.

	the exclusion criteria.	
	[The type of support sought from physicians and other sources of support] are examples of aspects of the study that were not incorporated in the aim stated at the end of the introduction. The aims need to include	The type of support sought and other sources of support were excluded from the objectives of this study.

	all aspects of the study, and these aspects need to be incorporated in the study rationale established in the introduction.	
	Study reference 53 is listed as a study from Pakistan in the reference list, but South Africa is stated here.	The references were corrected.
	Was there a reason for selecting the year 2000, or was this an arbitrary selection? Given the low number of studies prior to 2000, I am not convinced this sub-analysis adds much.	The comparison of studies by year of publication was removed from the objectives of this study.
	I am not sure there is sufficient sample sizes to support the majority of comparisons [of different sources of support about AAS]	Comparisons between physicians and other sources of support were excluded from the objectives of this study.
	It would be good to see some discussion of why ranges varied so widely in most countries. Presently, this paragraph mostly just expands on the relevant results rather than discussing why this may be the case. Likewise, what might explain the higher and less diverse prevalence rates in Australia compared to other countries/regions?	A discussion of barriers in the access to health services and possible explanations for the different prevalence rates were added to the article.
	Reviewer 3: Ms Adriane Lewin	
	In the opinion of this reviewer, all you need to present is a single number, pooled from all your included studies, which represents the % (or proportion – and please be clear in the methods and results that % and proportion are different) of AAS users who seek physician support. Then you can present pooled estimates from the subgroup analyses.	The study was re-written with greater focus on the overall prevalence and pooled estimates of the subgroup analyses, as suggested. The distinction between percentage and proportion was observed, and only percentages were reported.
	I am concerned and curious that the authors have chosen not to present any Forest plots.	The forest plots for overall and subgroups' effect sizes were included.

	I am uncertain why there is such a great emphasis on the Q statistic and why so many sub-group versus sub-group comparisons are presented.	The emphasis on Q statistics was removed from the study.
	You need to describe your methods in enough detail that another researcher could replicate your study and reach the same results	The methods were described in further detail and organised to increase the reproducibility of our study. The search algorithms were presented in the supplementary material, adapted for each database. The meta-analysis was performed in R and the full codes were shared in the supplementary material.
	Please assess publication bias.	The assessment of publication bias was included.
	Two reviewers should independently assess each paper for inclusion/ exclusion.	A new search was performed. Two independent reviewers screened and assessed the inclusion of papers, performed data extraction and assessed risk of bias.
	Section 2.2 does not describe data synthesis	The processes of data synthesis were described in the Methods section, after the assessment of risk of bias.
	If you do proceed with a meta-regression, you need to provide enough information here to allow another researcher to replicate your study: model building, model assumptions, how variables are categorised or coded, reference groups, dummy variables	The models and methods of the meta-regression were described, including the R codes that allow other researchers to reproduce the analysis.
	The results should lead with Figure 1 – the flow chart with inclusions and exclusions showing how your reached your study sample.	The flowchart and the selection of studies were moved to the beginning of Results.
	Table 2 gives characteristics of the included studies but I don't see any synthesis (analysis) here. Which is okay – just rename it to reflect the contents. Also, including multiple pieces of information in a single cell/column makes it difficult to	Table 2 was named 'Summary characteristics of selected studies' and the information was reorganised to facilitate the reading.

	read;	

	Data informing risk of bias should appear at the very least in its own table, and I also like to see a traffic light plot like used by Cochrane.org to present this information (for ease of assessment by readers).	The risk of bias was moved to eTable 3, including a traffic light plot.
	I question whether it is important to make the comparison [between the number of AAS users who seek physicians and other sources of support].	Comparisons between physicians and other sources of support were excluded from this study.

VERSION 2 – REVIEW

REVIEWER	Fink, Julius Juntendo University
REVIEW RETURNED	15-Mar-2022

GENERAL COMMENTS	According to the text, only 1% of the data comes from Australia, but the authors conclude from this small % that Australia has the highest prevalence of AAS seeking medical support. Is the data strong enough to make this statement hold true for the whole population? Why did the authors create an NSP category? The availability of NSP can vary a lot depending on the country. Is NSP something very popular amongst AAS users in the UK and Australia? Taken together, it seems like this study is biased by the Australian and UK data, which only represent a small fraction of the real population using AAS. I would suggest reevaluating your categories. Could the authors include a table representing the prevalence of medical advice and education/ and or financial status?

REVIEWER	Boardley, Ian University of Birmingham, Statistics, rehabilitation
REVIEW RETURNED	30-Mar-2022

GENERAL COMMENTS	I am happy with the revisions made in response to my original comments, and now believe the manuscript is ready for publication.

REVIEWER	Lewin, Adriane UNSW, Medicine/Whitlam Orthopaedic Research Centre
REVIEW RETURNED	24-Mar-2022

GENERAL COMMENTS	Steroids round 2 (Methods and Results only) Much better! The paper is now much easier to read and easier to follow. I did not receive the forest plots but would be happy to look at them and provide feedback. Bravo and thank you for including your R code and data – wonderful to see!! I recommend removing the entire meta-regression; it doesn't provide any additional information as you are assessing the effect of your secondary outcomes on the primary outcome. It is circular and doesn't make sense (at least to me). It would be different if, for example, you meta-regressed to assess the relationship between sample size, publication year, study type etc on the outcome. But you have already reported on the prevalence of AAS by location and user-types as secondary outcomes – let that be enough information. ps: I've uploaded a .doc and .pdf to which I added page and line numbers for ease of communication while reviewing. Specific comments: METHODS: • line 213 – provide the name of the supplementary table or figure with the search algorithm, e.g.(Supplementary table S1). • lines 214-216 and lines 222-223 - repeated phrase "Two independent reviewers (JA and AK) performed the data extraction" • line 467 - Please mention & reference the method used to assess publication bias • line 560 – I think this is an error as you don't include 'type

	of support sought' in this version. • line 554 – You can remove the meta-regression as it doesn't tell the reader anything new. • If you retain the meta-regression: line 564 – rather than 'contrast' use 'comparator' RESULTS: Overall: I'm not sure reporting any numbers to 2 decimal places is necessary – it implies a level of precision that is not present (what is 1 one-hundredth of one percent? Is it meaningful?) For all outcomes, add number of studies & number of participants for each subgroup, eg Australia (N studies; Y participants; estimate, 95% CI). Avoid the use of p-values entirely, with the exception of the primary outcome (and there, only if you want to). • line 590 – what is NSP? • line 771 – give the N here (rather than 'majority') • line 773 – 'regional' is duplicated • line 782 – ROB section – can you report n (%) of studies at high (ie not green) risk of bias in each of the 5 categories. • line 799 – in the text, just give the point estimate with 95% CI (don't need SE, P I2 here since all these data points should be included in Figure 3 [nb: I didn't receive any figures for this review so can't comment on them]). • line 866 – write out NSP in full (for your readers who have forgotten what this means and have to search through the paper to figure it out) • Line 887/Table 3 - I'm not convinced Table 3/ the meta-regression adds much to your paper and I'd be inclined to leave it out. You have already presented point estimates by location and subgroup in forest plots, which allow a reader to assess the data that feeds each of these estimates (the forest plots will show sample size and cases for each country, user type) and show visually the precision of each estimate. I recommend removing Table 3 and the meta-regression entirely. • If you do include Table 3 (but really, I don't think you should) I would strongly discourage use of a p-value anywhere in this paper, full stop. Include the point estimate and the 95% CI, and readers will be able to assess whether the point estimate is significantly different from zero or not. A p-value does not add much information, especially since your paper is largely descriptive (and the information is relevant and important and speaks for itself, without a p-value).

VERSION 2 – AUTHOR RESPONSE

	Reviewer 1: Dr Julius Fink	
	According to the text, only 1% of the data comes from Australia, but the authors conclude from this small %	An observation was added in the limitations of the study: The pooled prevalence in different locations was based

	that Australia has the highest prevalence of AAS seeking medical support. Is the data strong enough to make this statement hold true for the whole population?	on a limited sample of highly heterogenic studies. For instance, the estimated prevalence of AAS users seeking support from physicians from Australia was based on only four studies, comprising a total of 1,032 AAS users – 10.2% of the total number of AAS users from selected studies. We have acknowledged that this and other limitations can compromise the generalisation of our results.
	Why did the authors create an NSP category? The availability of NSP can vary a lot depending on the country. Is NSP something very popular amongst AAS users in the UK and Australia? Taken together, it seems like this study is biased by the Australian and UK data, which only represent a small fraction of the real population using AAS. I would suggest re-evaluating your categories.	Although only two studies with clients from the needle and syringe programme exchange (NSP) were included in this meta-analysis, a large body of literature have investigated the behaviour of AAS users who use the services of the NSP, namely in the UK and Australia. As previously observed by another reviewer of this journal, it was expected that the prevalence of AAS-using NSP clients seeking support from physicians would distort the overall prevalence rate (since this population was somehow engaged with the health system). Our data has partially corroborated this hypothesis: NSP clients showed the highest prevalence rate, but their pooled prevalence had no significant impact in the overall prevalence, as seen in the results of the meta-regression.
	Could the authors include a table representing the prevalence of medical advice and education/ and or financial status?	Very few of the selected studies have described the educational or financial status of their participants, compromising a comparison of this category of data.
	Reviewer 2: Dr Ian Boardley	
	I am happy with the revisions made in response to my original comments, and now believe the manuscript is ready for publication.	N/a

	Reviewer 3: Ms Adriane Lewin	
	I recommend removing the entire meta-regression; it doesn't provide any additional information as you are assessing the effect of your secondary outcomes on the	The purpose of our meta-regression is to measure the impact of 'location of study' and 'subpopulation of AAS users' on the overall prevalence rate of AAS users

	primary outcome. It is circular and doesn't make sense (at least to me). It would be different if, for example, you meta-regressed to assess the relationship between sample size, publication year, study type etc on the outcome. But you have already reported on the prevalence of AAS by location and user-types as secondary outcomes – let that be enough information.	seeking support from physicians. As discussed in the study, the meta-regression suggests that studies with adolescent AAS users have such an impact, due to their consistently low prevalence of seeking support from physicians. We believe that this information can be useful for future studies, where the help-seeking behaviour of adolescent AAS users should be taken into account. Besides, the meta-regression has shown that no particular location or other subpopulation of AAS users had a significant impact on the overall prevalence.
	METHODS line 213 – provide the name of the supplementary table or figure with the search algorithm, e.g.(Supplementary table S1).	The search algorithms were placed in eTable 1 of the supplementary material and the reference was changed accordingly.
	lines 214-216 and lines 222-223 - repeated phrase "Two independent reviewers (JA and AK) performed the data extraction"	This error was corrected.
	line 467 - Please mention & reference the method used to assess publication bias	The method used to assess publication bias was described and the references were added.
	line 560 – I think this is an error as you don't include 'type of support sought' in this version.	This error was corrected, and the sentence excluded.
	line 554 – You can remove the meta-regression as it doesn't tell the reader anything new.	We decided to keep the meta-regression, for the reasons explained above.
	If you retain the meta-regression: line 564 – rather than 'contrast' use 'comparator'	This was corrected.
	RESULTS I'm not sure reporting any numbers to 2 decimal places	We have decided to keep two decimal places, as it has been used in similar papers published by the BMJ (e.g.

	is necessary – it implies a level of precision that is not present (what is 1 one-hundredth of one percent? Is it meaningful?)	Panagioti et al., 2019. Prevalence, severity, and nature of preventable patient harm across medical care settings: systematic review and meta-analysis)
	For all outcomes, add number of studies & number of participants for each subgroup, eg Australia (N studies; Y participants; estimate, 95% CI).	The number of participants of each subgroup and the percentage they represent from the total of 10,101 AAS users investigated by the selected studies were added at the beginning of the Results.
	Avoid the use of p-values entirely, with the exception of the primary outcome (and there, only if you want to).	We decided to keep the p-values in the assessment of publication bias and on the meta-regression. We understand that the analysis of the 95% CI allow readers to verify if the tests are statistically significant, but we believe that the p value will help the readers to assess the significance faster, complementing the information of the 95% CI.
	line 590 – what is NSP?	The needle and syringe exchange programme, as described on the Introduction, line 197.
	line 771 – give the N here (rather than 'majority')	The number of studies was included.
	line 773 – 'regional' is duplicated	This was corrected.
	line 782 – ROB section – can you report n (%) of studies at high (ie not green) risk of bias in each of the 5 categories.	This description was added.
	line 799 – in the text, just give the point estimate with 95% CI (don't need SE, P I2 here since all these data points should be included in Figure 3 [nb: I didn't receive any figures for this review so can't comment on them]).	The SE, p and I^2 were removed from the text. The I^2 was added to the images.
	line 866 – write out NSP in full (for your readers who have forgotten what this means and have to search through the paper to figure it out)	As the meaning of NSP was shown at the Introduction, we decided to keep the abbreviation throughout the text.

	Line 887/Table 3 - I’m not convinced Table 3/ the meta- regression adds much to your paper	We decided to keep the meta-regression, for the reasons explained above.

	and I'd be inclined to leave it out. You have already presented point estimates by location and subgroup in forest plots, which allow a reader to assess the data that feeds each of these estimates (the forest plots will show sample size and cases for each country, user type) and show visually the precision of each estimate. I recommend removing Table 3 and the meta-regression entirely.	
	If you do include Table 3 (but really, I don't think you should) I would strongly discourage use of a p-value anywhere in this paper, full stop. Include the point estimate and the 95% CI, and readers will be able to assess whether the point estimate is significantly different from zero or not. A p-value does not add much information, especially since your paper is largely descriptive (and the information is relevant and important and speaks for itself, without a p-value).	We decided to keep the p-values on the meta-regression, as they inform if the impact of each variable was statistically significant.

VERSION 3 – REVIEW

REVIEWER	Fink, Julius Juntendo University
REVIEW RETURNED	26-Apr-2022

GENERAL COMMENTS	Just one thing I'd like the authors to confirm. In the discussion they state: ********** Two studies from German showed prevalence rates of AAS users seeking support from physicians of 37.97% [57] and 55.95% [49], while a single study from Poland reported a prevalence of 11.11% [53], despite the use of AAS not being illegal in both countries [81,82] ********** Are you sure that AAS are legal in Germany?

REVIEWER	Lewin, Adriane UNSW, Medicine/Whitlam Orthopaedic Research Centre
REVIEW RETURNED	14-May-2022

GENERAL COMMENTS	Thank you for the opportunity to review the updated manuscript. The manuscript needs additional description of the meta-regression and how it was conducted. Otherwise it is good, ready to go. Here are my comments: Methods: Page 8 line 8 – 'Variance' should be 'variability' Page 8, description of meta-regression needs more info : how did you build the model, how did you choose these two variables (location and population), why didn't you include other commonly used predictiors such as study quality or N? From the initial review: If you do proceed with a meta-regression, you need to provide enough information here to allow another researcher to replicate your study: model building, model assumptions, how variables are categorised or coded, reference groups, dummy variables, etc. For example, please describe how the candidate variables were chosen e.g. a priori based on research (cite reference). If they were selected post hoc, please mention this. Page 7 line 60 lists possible sources of heterogeneity "(e.g. different sample sizes, populations or methods)" - did you consider sample size or methods for the meta-regression? Please describe selection criteria for inclusion of variables in the final model, eg $p<0.1$ in univariable analyses. Page 11 line 41. A 4% reduction in heterogeneity from 98% to 94% is not meaningful and should be removed. In the Cochrane Handbook, heterogeneity between 75-100% is 'considerable'. 'strength' is misspelled throughout the paper.

VERSION 3 – AUTHOR RESPONSE

Reviewer 3: Ms Adriane Lewin		
	Page 8 line 8 – ‘Variance’ should be 'variability’	This was corrected.
	Page 8, description of meta-regression needs more info : how did you build the model, how did you choose these two variables (location and population), why didn’t you include other commonly used predictiors such as study quality or N? From the initial review: If you do proceed with a meta-regression, you need to provide enough information here to allow another researcher to replicate your study: model building, model assumptions, how variables are categorised or coded, reference groups, dummy variables, etc. For example, please describe how the candidate variables were chosen e.g. a priori based on research (cite reference). If they were selected post hoc, please mention this.	The meta-regression was refined, following the examples provided by the reviewer, the recommendations of Thompson and Higgins (2002) and similar studies (Sagoe et al, 2014; Panagioti et al, 2019). Following a previous recommendation of this reviewer, p values of the meta-regression were omitted. The meta-regression’s methods were further detailed and were transcribed below. The discussion of results was updated accordingly. Univariate and multivariate meta-regressions were performed to measure the impact of study level moderators on the prevalence of AAS users seeking support from physicians. Based on the variables utilised by Sagoe et al., four moderators were hypothesised to have an impact on the prevalence of AAS users seeking support from physicians (location of studies, subpopulation of AAS users, time of publication and study design). Additionally, two other moderators commonly used in prevalence studies were included post hoc (sample size and risk of bias). The selection and coding of moderators followed consensus procedures. The time of publication was categorised as before and after the year 2000, as we hypothesised that the availability of the internet and on-line support communities of AAS users – could impact the prevalence of AAS users seeking support from physicians. Risk of bias was categorised according to the number of negative or unknown responses to MMAT’s assessment criteria. For each moderator variable, the category with the highest number of studies was used as reference, and dummy variables were automatically generated. Statistically significant ($p < 0.05$) variables were entered into a multivariable model. The meta-analysis and meta-regression were

		conducted in R using the metaphor package. A full description of the codes and a dataset with the coded variables can be found in the supplementary material.
	Page 7 line 60 lists possible sources of heterogeneity “(e.g. different sample sizes, populations or methods)” - did you consider sample size or methods for the meta-regression?	This information was added (see above).
	Please describe selection criteria for inclusion of variables in the final model, eg p<0.1 in univariable analyses.	This information was added (see above).
	Page 11 line 41. A 4% reduction in heterogeneity from 98% to 94% is not meaningful and should be removed. In the Cochrane Handbook, heterogeneity between 75-100% is ‘considerable’.	This information was removed.
	‘strength’ is misspelled throughout the paper.	This was corrected.